

On the redundancy of rows

On-disk compression VS random-access compression

Marie Picard

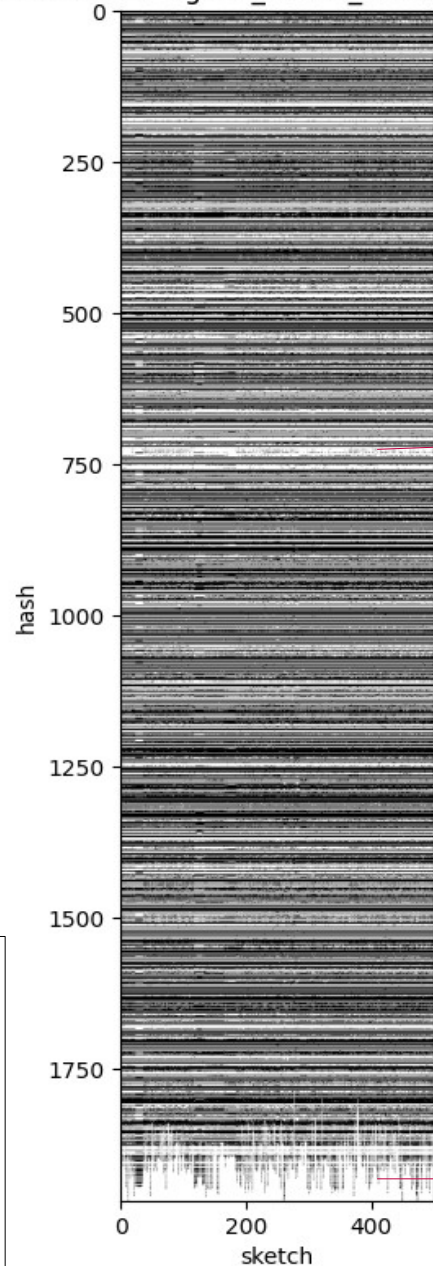
—

Supervisors : Karel Břinda, Leo Ackermann

January 23, 2026

Quick recap of past results

Presence-absence matrix for ngono_s1000_S500 - 500 sketches - s = 1000



Sparsity
comes
from
sparse
rows

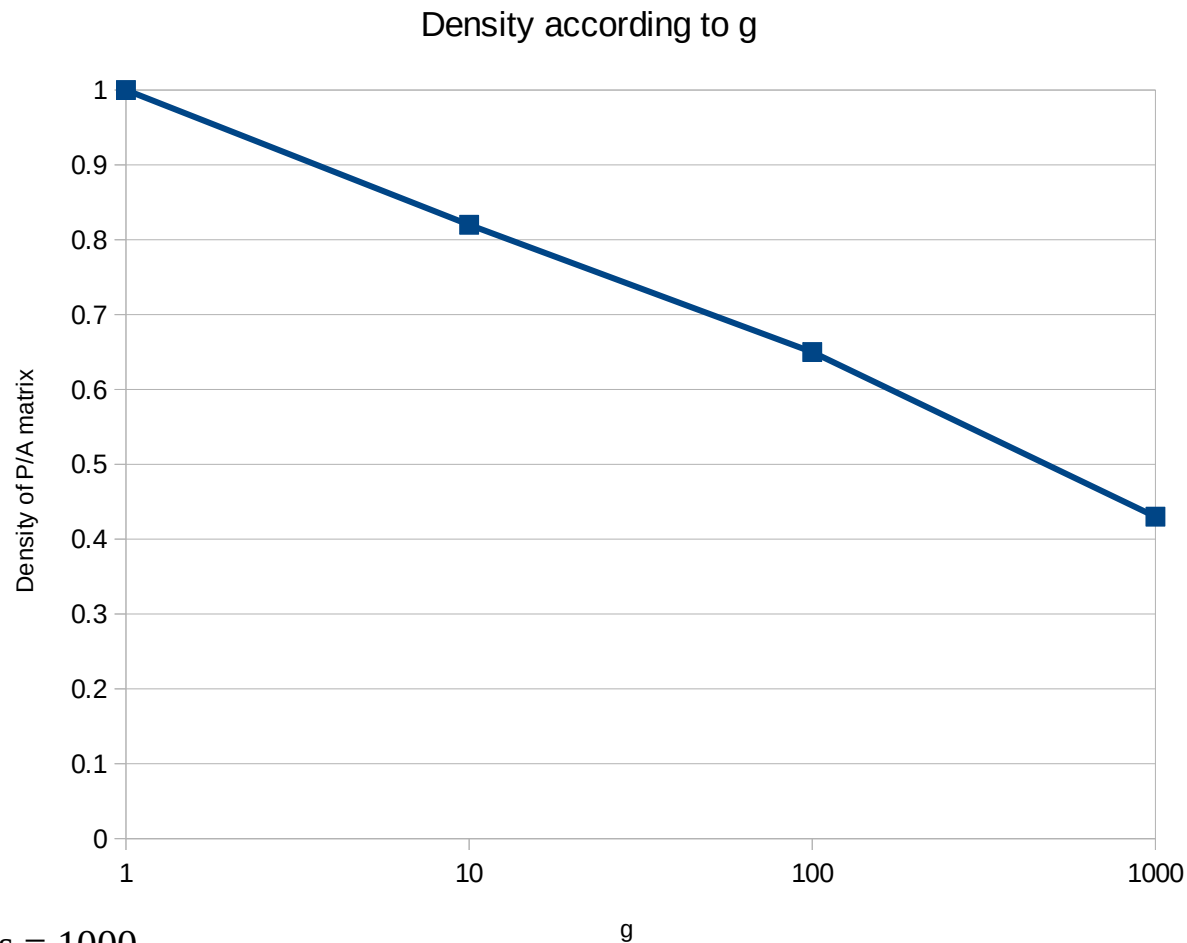
Small sparse area at the end

s	sketch size
g	number of genomes
δ	$\frac{1}{2} \sum_{i=1}^{g-1} d_{Hamming}(S(g_i), S(g_{i+1}))$
$\mathcal{N}_h$	number of distinct hashes
$\mathcal{S}_U$	union of all sketches
$\mathcal{M}$	presence matrix of hashes
d_1	density of ones in $\mathcal{M}$

- $g = 500$
 $s = 1000$
 • Density of matrix : 51%
 • Delta = 29.754

Quick recap of past results

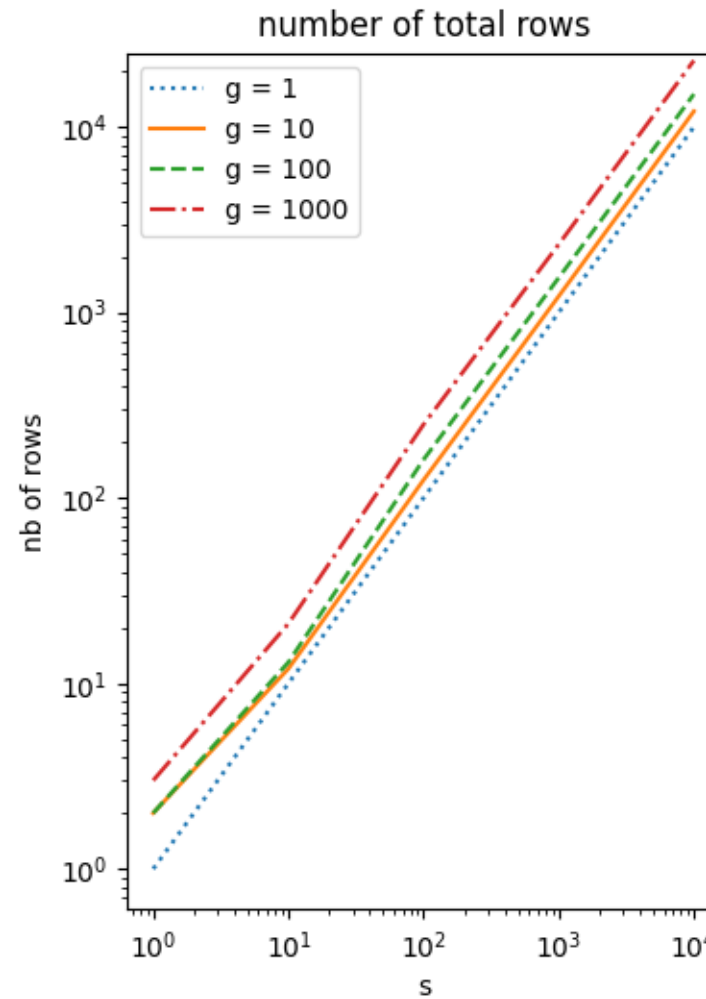
Density of ones in the presence/absence matrix



s	sketch size
g	number of genomes
δ	$\frac{1}{2} \sum_{i=1}^{g-1} d_{Hamming}(S(g_i), S(g_{i+1}))$
$\mathcal{N}_h$	number of distinct hashes
$\mathcal{S}_U$	union of all sketches
$\mathcal{M}$	presence matrix of hashes
d_1	density of ones in $\mathcal{M}$

- $s = 1000$
- *Neisseria gonorrhoeae*

Quick recap of past results



Goals

Reorder rows in the matrix to improve compressibility



ISSUE : breaking the ordering and Elias-Fano

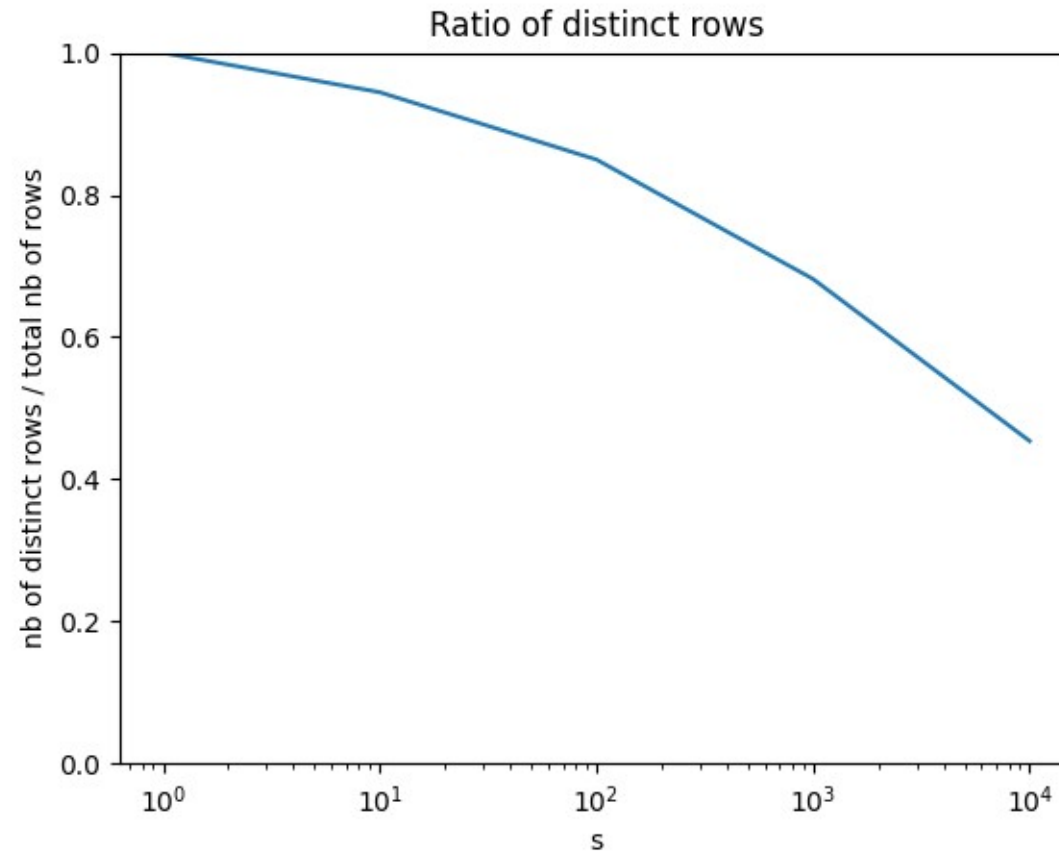
Split matrix in 2 :
dense and sparse
parts (still
ordered)

Remove
redundant rows

Experiment

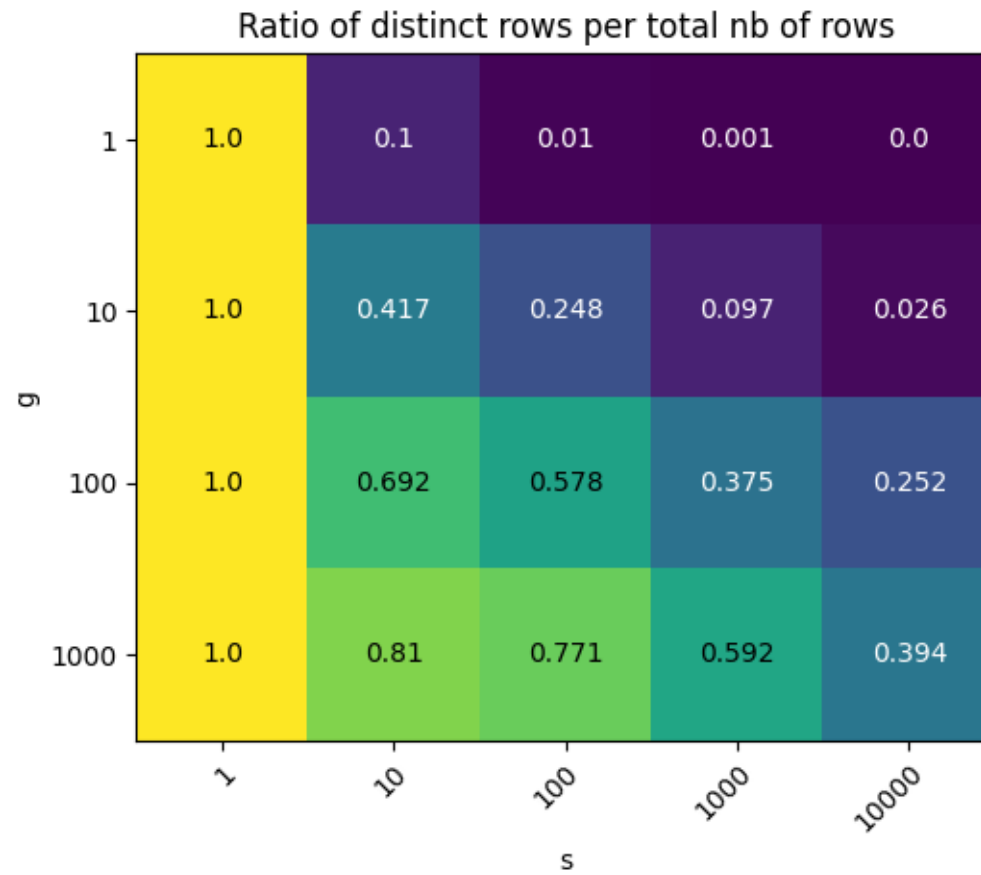
- g : number of genomes sketched (leftmost genomes)
- s : size of sketch
- Dataset :
neisseria_gonorrhoeae__01.tar.xz
<https://zenodo.org/records/15367750>

Results

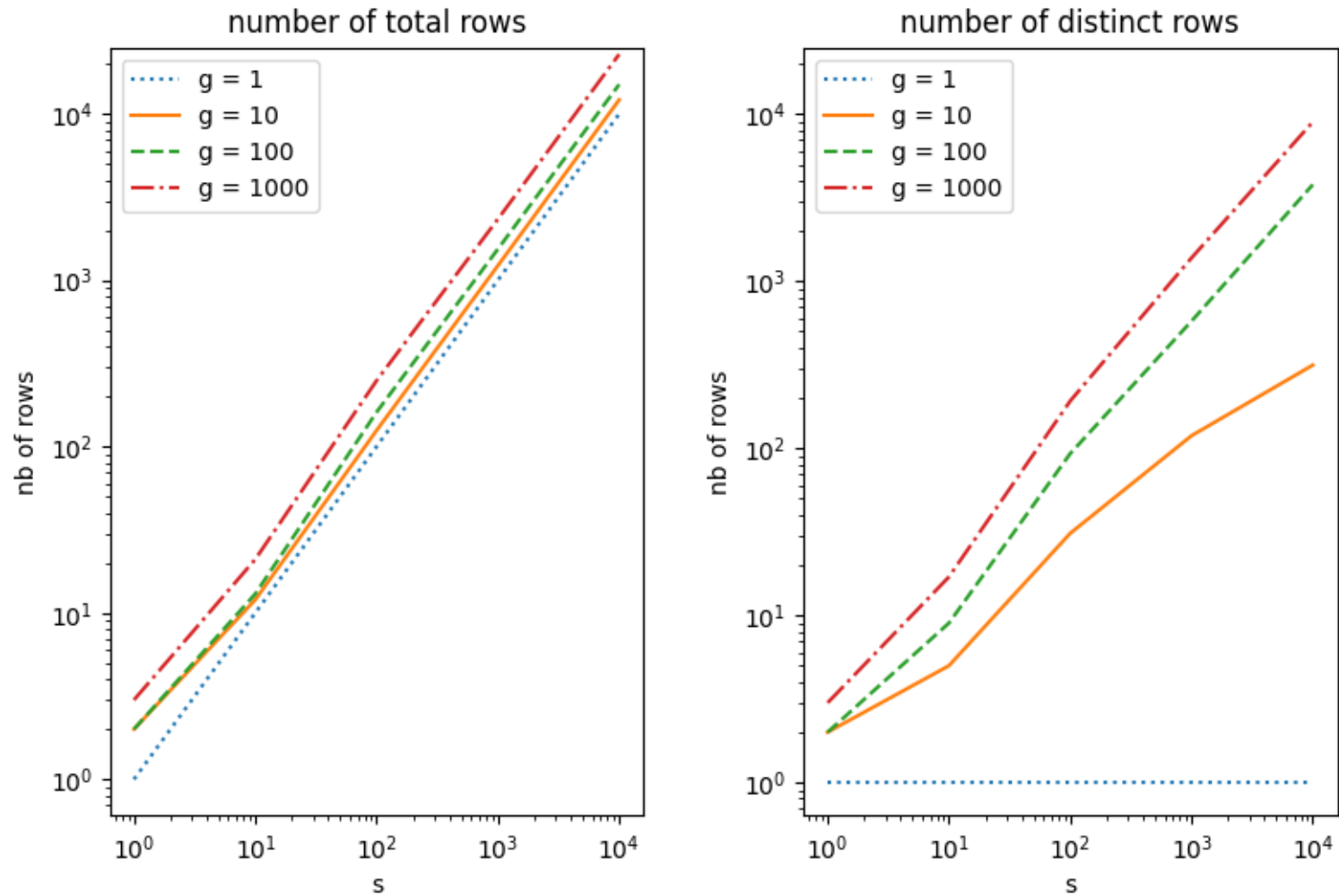


$g = 4000$

Results



Results



Conclusions

- Higher compressibility as s increases (limit : kmer matrices)
- Lower compressibility as the number of genomes increases (for redundant rows)
- Reordering rows : splitting matrix might be more efficient
- Look into XZ/GZIP/ZSTD compressibility

Extra stuff

- Easy operations on sketches (intersection : $\&$, union $|$, sketch : keep first s ones)
- RLE : look into Elias gamma/delta encoding for lengths
- RLE rank and select : Sunho Lee and Kunsoo Park
- Todo :
 - Implementation
 - Biological model